KAMALESH SURESH KUMAR

CYBERSECURITY ANALYST & PROTOCOL SECURITY RESEARCHER

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PROFESSIONAL SUMMARY

Computer Science undergraduate and security researcher with demonstrated practical experience in physical-layer and packet-level optical wireless security (Li-Fi), network vulnerability analysis, and penetration testing. Formally demonstrated custom exploit testing harnesses and protocol evaluations before the state Cybercrime Police Department. Strong command of network security protocols, Linux environments, applied cryptography, and disciplined operational defense.

TECHNICAL SKILLS & CORE COMPETENCIES

Security Toolchains: Optical Wireless Communication (OWC), Packet Analysis, Signal Processing
Languages: Python, C/C++, Bash / Shell Scripting, TypeScript, SQL
Operating Systems & Tools: Linux (Debian/Ubuntu/Arch), Git, Network Analyzers
Concepts: Encryption, Attack Surface Mapping, Air-Gapped / Eavesdropping Vectors

Key Competencies: Optical Wireless Security (Li-Fi RX 2.0/3.0) * Penetration Testing & Exploit Analysis * Network Protocol & Packet Analysis * Physical-Layer Vulnerability Assessment * Applied Cryptography & Key Exchange * Linux/Unix Security & Shell Hardening

KEY ENGINEERING PROJECTS

Li-Fi Penetration Testing & Optical Security Protocols | *Python, C/C++, OWC Signal Processing, Network Security, Linux*

- Researched, developed, and deployed experimental penetration testing protocols for optical wireless transmissions (Li-Fi RX 2.0 and RX 3.0).
- Conducted simulated eavesdropping, optical packet injection, and signal attenuation attacks in controlled laboratory testbeds.
- Demonstrated findings and mitigation strategies directly to cybercrime law enforcement specialists during formal technical briefings.

KAI HQ — Headquarter Operating System | *React 19, TypeScript, Client-Side Sanitization, Static Architecture*

- Engineered an entirely static, serverless digital headquarters eliminating server-side attack surfaces, SQL injection vectors, and unauthorized data leakage.
- Implemented clean routing and state encapsulation with zero external analytics tracking.

Automated CI/CD Security & Integrity Verification Pipeline | *GitHub Actions, Bash, TypeScript Compiler*

- Engineered deterministic CI verification gates preventing unvetted dependencies or failing lint rules from reaching production builds.

EXPERIENCE & LEADERSHIP

Naval Wing Cadet & Military Leadership — National Cadet Corps (NCC - Navy Wing) *(2022 — Present | India)*

- Underwent rigorous military discipline, naval operational procedures, crisis management, and field protocol execution.
- Developed high resilience, team command, and adherence to operational security under high-stress scenarios.

Software Development Intern — Viyuga Innovations *(2024 | India)*

- Contributed to secure application design, input validation, and code review processes.

Technical Trainee / Intern — Infosys SpringBoard *(2023 — 2024 | Virtual / India)*

- Completed comprehensive coursework on enterprise network models, cloud security fundamentals, and distributed computing.

EDUCATION

B.E. in Computer Science and Engineering — Undergraduate Engineering Program *(2022 — Present | India)*

- Core Coursework: Systems Architecture, Data Structures, Algorithms, Operating Systems, Computer Networks, Database Systems, Network Security.

CERTIFICATIONS & VERIFIED HONORS

- Li-Fi Penetration Testing Demonstration & Law Enforcement Briefing — Cybercrime Police Department (2024)
- Enterprise Software & Cloud Computing Certification — Infosys SpringBoard (2023 — 2024)
- NCC Naval Wing Cadet Leadership & Service — National Cadet Corps (2022 — Present)